# Raihan Rafeek

+1-513-580-9384 | rafeekrn@mail.uc.edu | linkedin.com/in/raihan-rafeek | github.com/Rai1975 | www.rai-1975.com

## Summary

Software engineer and AI/machine learning (ML) developer with internship experience in agentic AI systems, full-stack development, and data science. Built and deployed Large Language Model (LLM) agents and Retrieval-Augmented Generation (RAG) pipelines. Prize winner at 4 hackathons with 8 awards in total, including 2 national-level events with 900+ participants. Research experience in Topological Data Analysis (TDA) and algorithm runtime optimization with a co-authored paper.

## Technical Skills

**Languages:** Python, Ruby, Go, JavaScript, TypeScript, SQL, Cypher
**AI/ML:** Machine Learning, Deep Learning, Large Language Models (LLMs), Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), PyTorch, TensorFlow, scikit-learn, LangChain, CrewAI, ExaAI, Strands, Hugging Face Transformers, Sentence-Transformers, agentic AI, NumPy, Pandas, SciPy
**Cloud and DevOps:** AWS (Bedrock, Lambda, EC2), cloud hosting, Docker, containerized microservice deployment (Railway), Render, Cloudflare, Supabase
**Web and Backend:** Flask, Ruby on Rails, REST APIs, React, Next.js, Svelte, Material UI, Electron, Twilio
**Databases:** Microsoft SQL Server, PostgreSQL, Neo4j (graph database)
**Practices:** Full-Stack Development, Backend Development, Agile, query optimization, system design, Agentic Automation

## Professional Experience

**Product Engineer Intern** | Benchmark Gensuite Mar 2026 – Aug 2026

- Restructured and optimized API endpoints to support agentic tool interactions, improving AI agent compatibility while obfuscating PII using JWT-based authentication
- Designed and deployed a customer facing dynamically extensible agent architecture spanning 5+ product capabilities, enabling users to generate reports, manage tasks, and execute actions through a unified interface
- Built an internal AI-powered reporting, task-management and cost monitoring workflow, reducing user navigation from 4–5 pages to 1 interface
- Optimized agent orchestration and inference pipelines, reducing average response time by 75%

**AI and Data Science Intern** | Exelixi AI Jan 2025 – Apr 2025

- Developed machine learning models tailored to customer requirements using scikit-learn, PyTorch, and other ML frameworks
- Designed and implemented interactive dashboards visualizing model performance, key feature contributions, and cost-efficiency metrics for stakeholders
- Created and presented product demos at multiple conferences across the UAE, showcasing the company's projects, services, and long-term vision
- Collaborated cross-functionally in a fast-paced startup environment to deliver projects from concept to completion on schedule

**Undergraduate Research Assistant** | Experimental Computation Lab, University of Cincinnati Aug 2024 – Jun 2025

- Assisted in development of algorithms to extract features from high-dimensional datasets using Topological Data Analysis (TDA), including persistent homology
- Implemented dimensionality reduction techniques (PCA, t-SNE, UMAP) to reduce dataset complexity while preserving critical features for downstream tasks
- Reduced time complexity and memory usage to enable faster processing of larger datasets
- Co-authored paper: Topological Study of $\beta$-Sparsified $d$-Uniform Hypergraph Based Simplicial Complexes

**Full Stack Software Engineer Intern** | Materials and Fuels Complex, Idaho National Laboratory Aug 2023 – Dec 2023; May 2024 – Aug 2024

- Developed safety-critical nuclear inventory management software using Ruby on Rails and PostgreSQL in compliance with strict regulatory standards
- Improved search result filtering speed by optimizing queries through a database migration and implementing caching
- Resolved 30+ bugs, shipped new features, and fixed UI/UX inconsistencies to ensure reliable application behavior
- Collaborated with engineers in an Agile environment and received mentorship from experienced professionals

## Research and Technical Projects

**ADCS Physics Simulation** | LEOPARDSAT-1 CubeSat Team, University of Cincinnati CubeCats (partnering with NASA and Nanoracks) Aug 2024 – Oct 2025

- Built simulation software for Passive Attitude Determination and Control Systems (ADCS) using Pandas, NumPy, and

SciPy
- Applied physical models of magnetic materials and hysteresis curves to simulate magnetic torques within the CubeSat detumbling algorithm, improving attitude control efficiency and stability
- Integrated a data pre-processing pipeline for a synthetic dataset of 6,000,000+ data points to reduce hysteresis model runtime
- Collaborated with interdisciplinary team of electrical, aerospace and mechanical engineers to align simulation models with mission objectives

## Projects

**Randy the Otter | 2026** | Docker, LaTeX, Railway
- Built Randy the Otter, an AI job-hunt assistant (Chrome extension + AWS Strands Agents/Bedrock orchestrator) that scrapes job postings via the DOM, scores candidate fit, autofills applications, and tracks applied jobs
- Generated tailored resumes and cover letters via a Docker-based LaTeX compiler microservice hosted on Railway

**Whitebox | 2024** | Python, Flask, React, Neo4j, LangChain, Llama, Hugging Face, OpenAI, Docker
- First place winner, Future of Data Hackathon at Midwestcon (900+ participants, 70+ teams)
- Transformed credible texts such as medical textbooks into structured knowledge graphs, enabling explainable AI reasoning that users can trace step by step

**Rent Radar | 2025** | Python, Flask, LangChain, React, Material UI, PostgreSQL, Supabase, Render, Cloudflare
- Won 2nd place overall at the CEAS Summer Innovation Challenge
- Deployed live platform with 50+ active users that crowdsources honest rental property information to increase landlord accountability
- Secured funding to operate Rent Radar as a startup, launching on-campus campaigns that grew the platform to 1,000+ crowdsourced rental reviews

**Memento | 2024** | Python, Flask, React, PostgreSQL, OpenAI
- Won 3rd Place Overall, Best Social Impact Hack, Best Use of Tembo, and the MedPace Award at RevolutionUC
- Built a journaling app for Alzheimer's and dementia patients and caretakers, with a RAG-powered assistant that answers questions grounded in journal entries to limit hallucination

**Gyrus | 2025** | Python, Flask, Electron, Neo4j, CrewAI, LangChain, Hugging Face, Sentence-Transformers
- Built an agent-based browser using an ensemble of CrewAI and LangChain agents with intent-memory mapping to anticipate user intent and direct users to relevant sources

**MindLine | 2026** | Python, Twilio, Flask, Strands, React, Material UI, PostgreSQL
- Built MindLine, a smart journaling platform that uses a spaCy NLP and sentiment/emotion pipeline to extract people, places, and actions from entries, and an agentic pattern-discovery system linked to a knowledge graph to surface personalized insights; won 2nd Place at the 2026 CEAS Summer Innovation Challenge

**Hi-Five | 2024** | Python, BERT, OpenAI, Hugging Face, Next.js, Neo4j
- Built a matchmaking app using 5-dimensional OCEAN personality embeddings combined with user interests to connect people with similar personalities

**Terminally-Addicted | 2024** | Python, Go, curses
- Won Most Creative Hack at HackOH/IO
- Built a CLI productivity app integrating Spotify, GitHub, ChatGPT, and Todoist, with in-terminal video playback

## Leadership and Activities

**Project Lead** | NEXT Innovation Scholars, University of Cincinnati Aug 2025 – Present
- Led stakeholder interviews with U.S. Bank partners to identify opportunities to strengthen the partnership between U.S. Bank, 1819 Innovation Hub, and UC
- Conducted stakeholder interviews and insight generation with UC's College of Cooperative Education and Professional Practice on the future of experiential education and student-employer partnerships
- Conducted landscape research and foresight-based world-building to develop plausible future scenarios for society, technology, and institutions

## Education

**University of Cincinnati** | B.S, Computer Science | May 2027 | GPA: 3.98

**Relevant Coursework:** Intelligent Systems (CS5136), User Interface Design (CS5167), Topological Data Analysis, Machine Learning
- Intelligent Systems: Implemented perceptrons (97% accuracy on MNIST), feed-forward networks, auto-encoders, and self-organizing feature maps from scratch without ML frameworks